H.M.R.L.B HERATH

Rajitha

Mechatronics Engineer · Lecturer · Robotics Researcher
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PROFILE

Mechatronics engineer and university lecturer with research experience in bio-inspired underwater robotics, CFD-based hydrodynamic analysis, mechanism design and experimental validation. Completed an MPhil at Shandong University of Science and Technology, where my research focused on developing and evaluating a multi-modal underwater robot capable of different propulsion mechanisms. Currently teaching embedded systems, microprocessor architecture and mechatronic design while supervising undergraduate engineering projects.

Alongside this I have served as a university instructor, level coordinator, team captain, competitive swimmer and award-winning performer combining deep technical skill with leadership, teaching experience and creativity. I bring analytical thinking, teamwork, fast learning and innovation to any engineering role.

EDUCATION

Master of Philosophy (MPhil) in Mechanical and Electronic Engineering *Sept 2022 – June 2025*
Shandong University of Science and Technology, Qingdao, China

- **Specialisation:** Robotics Engineering — Underwater Bio-Inspired Robotics
- **Thesis:** "Numerical Simulation and Experimental Research on the Hydrodynamic Characteristics of Multi-Modal Underwater Robots"
- **Supervisor:** Assoc. Prof. Zhang Minghui (张明辉)
- One semester completed online owing to COVID-19 restrictions.

Bachelor of Engineering (BEng) in Mechatronic Engineering *2022*
Shandong University of Science and Technology, Qingdao, China

- **GPA 4.23 / 5.00 /** Average grade 91.8

CS50: Introduction to Computer Science — Certificate
HarvardX

ACADEMIC APPOINTMENTS & PROFESSIONAL EXPERIENCE

Lecturer — General Sir John Kotelawala Defence University (KDU) *May 2026 – present*
Level Coordinator

- Teach undergraduate modules in embedded systems, microprocessor architecture and mechatronic design.
- Develop laboratory sheets, assignments and marking schemes for the microprocessor module.
- Designed and deployed browser-based teaching simulators for the Intel 8085 microprocessor and the STM32 NUCLEO-F411RE board, used in laboratory delivery and for self-paced student practice.
- Supervise individual projects and final-year projects.
- Conduct technical workshops preparing students for IMechE engineering competitions.
- Contribute to the organisation of ROSCO'26, an inter-university robotics competition, including the design of its supporting systems.

Visiting Lecturer — SIBA Campus *Jan – June 2026*
Department of Information and Technology

Instructor — Sri Lanka Institute of Information Technology (SLIIT), Sri Lanka *Aug 2022 – Jan 2023*

- Conducted weekly assessments for the RC UAV project with 2nd-year Mechatronics Engineering students.
- Delivered practical classes on Automation for 4th-year Mechatronics undergraduates, and Advanced Automation for 2nd-year Electrical Plant students.
- Organised and led a workshop on BCP Design and practical sessions on PD Control for 2nd-year Mechatronics Engineering students.
- Taught Automation programming to 3rd-year Electrical and Electronic Engineering students (BEng Hons — LJMU).

Trainee Engineer — SriLankan Airlines *July – Oct 2021*

- Gained hands-on experience in aircraft maintenance operations and aviation safety standards.
- Led the design and optimisation of a Toilet Assembly Test System for aircraft lavatories.

International Student Volunteer — Shandong University of Science and Technology, China *2023 – 2024*

PUBLICATIONS

PEER-REVIEWED CONFERENCE PAPER

Zhang Minghui & **Herath Rajitha Lakmal Bandara** (2024). "Research on Motion Mechanism of a Snake-like Robot in Water Environment." *Proceedings of the 2024 3rd International Symposium on Control Engineering and Robotics*, Changsha, China, 13 August 2024. Published in the ACM Digital Library. doi:10.1145/3679409.3679497

THESIS

Herath Rajitha Lakmal Bandara (2025). "Numerical Simulation and Experimental Research on the Hydrodynamic Characteristics of Multi-Modal Underwater Robots." MPhil thesis, Shandong University of Science and Technology, Qingdao, China.

RESEARCH & ENGINEERING PROJECTS

Bio-Inspired Underwater Hybrid Robot ***MPhil research project***
Designed and developed a bio-inspired multi-arm underwater robot with a mechanism capable of switching between different propulsion modes. Investigated its hydrodynamic performance using CFD simulations and experimental validation, with particular focus on propulsion efficiency and underwater locomotion.

Robotic Arm Design *MPhil preparation course*
Designed and developed a robotic arm, focusing on mechanical structure, actuator selection and control integration for precise movement — demonstrating robotics principles, system integration and CAD modelling.

System to Increase the Efficiency of a Solar Panel System ***Undergraduate final-year project***
Designed and implemented a system that tracks solar intensity and dynamically adjusts panel orientation to optimise energy output.

Toilet Assembly Test Setup ***SriLankan Airlines, Avionics Dept.***
Designed and built a custom test rig supporting the maintenance and quality assurance process for aircraft toilet assemblies.

Mechanical Drive Design for a Screw Conveyor
Engineered a mechanical drive system to transport slurry efficiently at a construction site, accounting for torque requirements, material build-up and environmental conditions.

PIR Sensor Alarm System
Developed a motion-activated alarm system using Passive Infrared sensors for low-cost security applications.

Gearbox Speed Reducer Design ***Undergraduate CAD project***
Designed a mechanical gearbox speed reducer in PTC Creo, focusing on gear arrangement, housing and 3D assembly.

TECHNICAL EXPERTISE

- **Robotics & Mechatronics —** mechanism design, actuation systems, tendon-driven mechanisms, robotic arms
- **Simulation & Analysis —** ANSYS Fluent (CFD), dynamic grid motion, MATLAB
- **Control Systems —** PID control, motion control, actuator tuning, dynamic systems
- **Embedded Systems —** ESP32, Arduino, microcontroller programming, sensors, IoT systems
- **CAD & Design —** SolidWorks, PTC Creo, Fusion 360, mechanical design, tolerancing
- **Electronics —** PCB design, signal processing, relays, automation circuits
- **Software & Programming —** Python, C, MATLAB, Assembly language, ANSYS programming
- **Home Automation & IoT —** smart home systems, Wi-Fi control, sensor and relay integration

Leadership & professional skills — leadership and team management; teaching and technical communication; research and innovation; cross-cultural communication; problem-solving and adaptability; project management.

AWARDS, SCHOLARSHIPS & HONOURS

ACADEMIC & RESEARCH

- **Chinese Government Scholarship (CSC) —** full scholarship for all years of master's study, Shandong University of Science and Technology.
- **Outstanding Student Award, Master's Programme (2025) —** Shandong University of Science and Technology, Qingdao, China. Awarded for excellence in academic performance and active involvement in extracurricular activities and university events during postgraduate studies.
- **Shandong Provincial Government Scholarship —** awarded for three consecutive years during undergraduate studies for academic excellence.

SPORT

- **SLUSA Colours —** Sri Lanka University Sports Association (2017), awarded for representing Sri Lanka at the Universiade Games (World University Games) in Taipei.
- **Provincial Colours, Central Province, Sri Lanka —** awarded for outstanding performance in swimming, representing the province at national-level competitions.
- **High School Colours, Kingswood College, Sri Lanka —** awarded for excellence and leadership in swimming during secondary education.

ARTS & CULTURE

- **Graduate Singing Competition, 2024 —** Second Place and the Award for Best Original Song Creation.
- **"Mountain and Sea Green Connection — My Story of Studying Qingdao" Competition, 2024 —** First Prize.

LEADERSHIP, SERVICE & COMMUNITY ENGAGEMENT

Student Councillor & Volunteer *2022 – 2024*
Shandong University of Science and Technology, China

- Volunteered in campus-wide events and cross-cultural initiatives.

Graduate Union Volunteer — Outreach Programme *2022 – 2025*
Shandong University of Science and Technology, China

- Participated in educational outreach visits to rural areas of China.
- Engaged with school students, shared personal academic experiences and contributed to their learning and motivation.

Media Representation

- Represented SDUST on Qingdao TV and the BERIS Translation Centre on Mango TV (China), showcasing talents and sharing experiences as an international student in China.

Sport & Society Leadership

- Member of the Sri Lankan contingent to the 29th Summer Universiade, Taipei (2017) — represented Sri Lanka in international competition.
- Member, Central Province Swimming Team, Sri Lanka (2011, 2012, 2013, 2016) — competed at national level.
- Captain, Kingswood College Swimming Team (2013 – 2014).
- Coordinator, Western Music Society, Kingswood College, Kandy (2011 – 2012).

REFEREES

Prof. Kapila C.K. Perera
Senior Professor of Mechanical Engineering, University of Moratuwa
Secretary, Ministry of Transport, Highways, Ports and Civil Aviation, Sri Lanka
Former Vice Chancellor, University of Moratuwa; former Secretary, Ministry of Education, Sri Lanka
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Prof. Nimal Punyasiri
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Assoc. Prof. Minghui Zhang *— MPhil supervisor*
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